

Cross-validating an open-source Rust FEM solver against Palace on a superconducting transmon geometry

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Abstract

Palace, the open-source finite-element electromagnetics solver from the AWS Center for Quantum Computing, has become a reference tool for superconducting-qubit design, yet no independent quantitative validation of it exists in the literature. We supply one. GEODE-FEM, an independently implemented open-source Rust finite-element solver, reproduces Palace’s eigenmode solution on a real transmon-plus-readout-resonator geometry — generated by DeviceLayout.jl’s `SingleTransmon` example and meshed once, so both solvers consume the identical mesh and the identical lumped-element junction model — to within 0.033% across all six modes (worst case 0.032%), with the junction LC mode agreeing to 0.001%. Exact analytic tripwires (the LC anchor and the Josephson $1/\sqrt{2}$ inductance scaling) guard against coincidental agreement. On an 8-core CPU node, geode-fem’s serial direct-factorization shift-invert eigensolve is roughly $4\times$ more efficient per core than Palace’s MPI-distributed Krylov solve at this 133k-DOF size, while Palace’s parallelism wins $1.7\times$ at the whole-node level — an architecture trade-off, not a ranking. A first GPU cell establishes CUDA-f32 correctness for geode-fem’s driven-solve kernels on a physical NVIDIA L40S; GPU performance measurement is explicitly deferred, and no GPU performance claim is made. Two caveats are stated up front: the model contains no ~ 4 GHz qubit mode *by construction* (both solvers agree on its absence), and geode-fem’s un-projected Lanczos admits one disclosed spurious junction-localized mode that Palace’s divergence-free projection suppresses. The full fixture-to-comparison pipeline is scripted and committed.

1 Introduction

Three-dimensional finite-element electromagnetics is now a load-bearing step in superconducting-qubit design: mode frequencies, dispersive shifts, and participation ratios extracted from field simulation feed directly into device Hamiltonians (Koch et al., 2007; Mineev et al., 2021a; Nigg et al., 2012). The open-source solver of record for this workflow is Palace (Palace developers, AWS Center for Quantum Computing, 2023), an MFEM-based (Anderson et al., 2021; Andrej et al., 2024) parallel solver from the AWS Center for Quantum Computing, typically driven by geometry from DeviceLayout.jl (AWS Center for Quantum Computing, 2025; Peairs and Carson, 2025). This stack is widely used and well engineered — and, unusually for scientific software of its influence, has never been the subject of a peer-reviewed or preprint publication: Palace exists as a repository, a blog post (AWS Quantum Technologies Blog, 2023), and a workshop talk; the DeviceLayout.jl transmon workflow exists as a blog post and a conference talk (Peairs, 2025; Peairs and Carson, 2025). Independent, quantitative, third-party validation of the stack does not exist in the literature.

This paper supplies that validation, and does so under conditions designed to make agreement meaningful rather than merely reassuring. We compare Palace against GEODE-FEM (“geode-fem”), an open-source finite-element solver implemented independently in Rust — different language,

different assembly path, different eigensolver, different linear algebra — on a real transmon-plus-readout-resonator geometry produced by DeviceLayout.jl’s own `SingleTransmon` example. The geometry is meshed *once*; both solvers consume the identical mesh, the identical first-order discretization, and the identical lumped reactive-shunt junction model. Agreement claims are therefore about formulation and solver correctness, not meshing luck.

Our contributions are:

- **The first independent validation of the Palace/DeviceLayout.jl transmon stack.** On the identical mesh at matched order, geode-fem reproduces all six of Palace’s eigenmodes to $\leq 0.033\%$ (worst case 0.032%), with the junction LC mode agreeing to 0.001% (Section 6).
- **A reproducible same-mesh cross-solver methodology.** An oracle-first benchmark protocol — exact analytic tripwires, inverse tests that must fail, participation-based mode identification, and a fully scripted fixture-to-comparison pipeline (Sections 5 and 11).
- **An honest architecture-level performance comparison.** On an 8-core node, a serial direct-factorization shift-invert eigensolve is $\sim 4\times$ more efficient per core than a distributed Krylov solve at 133k DOFs, while MPI parallelism wins $1.7\times$ at the whole-box level (Section 9); a GPU cell establishes driven-path CUDA correctness on physical hardware and defers its performance study until measured artifacts exist (Section 10).
- **Honest-negative reporting as a first-class result.** The model’s lack of a ~ 4 GHz qubit mode is explained and confirmed by both solvers (Section 7); a spurious mode leaked by geode-fem’s un-projected Lanczos is disclosed, characterized, and tied to a known fix (Section 8).

Section 2 positions this work; Section 3 defines the benchmark problem; Section 4 contrasts the two solver architectures; the remaining sections present methodology, results, and reproducibility details.

2 Related Work

Electromagnetic simulation workflows for superconducting qubits. The closest prior work consists of two recent Palace-based workflow papers. Sommers et al. (2025) present SQDMMetal, an open workflow that couples Qiskit Metal, Gmsh, and Palace, and cross-check Palace against the commercial solvers COMSOL and Ansys HFSS on qubit and resonator geometries, reporting agreement at roughly the $0.02\text{--}0.13\%$ level. Ye et al. (2025) build a Palace-centered layout-to-Hamiltonian workflow and benchmark it against cryogenic measurement, with resonator frequencies reproduced to within 0.3% . Both efforts treat Palace as a component to be *wrapped and trusted*: validation runs against commercial tools or experiment, through independently constructed models and meshes, so residuals mix solver error with meshing and model-entry differences. Neither is a same-mesh, same-formulation comparison against an independently *implemented* solver. Our study eliminates meshing and model entry as confounders — identical MSH fixture, identical lumped-shunt junction, first-order Nédélec elements on both sides — so residuals isolate formulation and solver correctness; to our knowledge it is the first independent third-party validation of Palace itself. The broader open ecosystem for qubit electromagnetic design includes Qiskit Metal and its quasi-lumped analysis methods (Minev et al., 2021b) and the SQuADDS validated design database (Shanto et al., 2024); the physics framing for extracting qubit parameters from field modes rests on the transmon (Koch et al., 2007), black-box quantization (Nigg et al., 2012), and energy-participation-ratio (Minev et al., 2021a) literature.

Verification, validation, and cross-solver benchmarking. Code-to-code comparison on precisely specified benchmark problems is an established genre in computational electromagnetics: the TEAM (Testing Electromagnetic Analysis Methods) workshop tradition has run such comparisons since 1986.¹ In the V&V vocabulary of Roache (1997) and Oberkampf and Roy (2010), the present study is verification-side evidence: our analytic tripwires are code-verification instruments in exactly Roache’s sense, and the same-mesh protocol is a code-to-code comparison in Oberkampf and Roy’s taxonomy. Multi-solver comparisons on shared electromagnetic problems have precedent in nano-optics, where Hoffmann et al. (2009) compared FEM, FDTD, and FIT codes on plasmonic antennas. We import that discipline to the superconducting-qubit domain and add two elements the genre often lacks: exact analytic tripwires that must move in known ways, and honest-negative reporting of what the model does *not* contain.

Open-source FEM/EM solver ecosystems. Mature open electromagnetics tools are normally accompanied by a citable methods paper: GetDP (Dular et al., 1998), DOLFIN/FEniCS (Logg and Wells, 2010), Gmsh (Geuzaine and Remacle, 2009), and scikit-rf (Arsenovic et al., 2022) all have one; so does Palace’s substrate library MFEM (Anderson et al., 2021; Andrej et al., 2024). Palace and DeviceLayout.jl are outliers — the *library* is published, the solver product built on it is not — which is precisely the gap this benchmark narrows from the outside, at a formality level (preprint) exceeding the original stack’s blog-and-talk record.

Numerics. geode-fem’s eigenpath is a shift-invert Lanczos (Lehoucq et al., 1998) over a real symmetric pencil, factorized with the faer sparse-LU (Kazdadi, 2026) and assembled on the Burn tensor framework (Tracel AI, 2024); the element family is first-order Nédélec (Nédélec, 1980). Palace’s eigenpath is a SLEPc-class Krylov method (Hernandez et al., 2005) with divergence-free projection and auxiliary-space (AMS) preconditioning (Hiptmair and Xu, 2007; Kolev and Vassilevski, 2009).

3 Benchmark Problem: Geometry, Mesh, and Junction Model

3.1 Geometry and mesh provenance

The device is the `SingleTransmon` example shipped with DeviceLayout.jl v1.15.0 (AWS Center for Quantum Computing, 2025): a transmon with readout resonator on a sapphire substrate, enclosed in a vacuum box. The sapphire is anisotropic, with permittivity tensor $\varepsilon = R \text{diag}(9.3, 9.3, 11.5) R^T$ rotated approximately 36.87° in-plane. The DeviceLayout.jl workflow emits seven named physical groups (metal, substrate, vacuum, junction lumped-element surface, two readout ports, exterior boundary) and drives Gmsh (Geuzaine and Remacle, 2009) to produce a tetrahedral mesh, exported as an MSH 4.1 fixture and pinned by SHA-256 (Section 11).

The mesh has 22,684 nodes and 133,314 tetrahedra, yielding 156,863 Nédélec edge unknowns, of which 133,108 interior degrees of freedom remain after eliminating perfect-electric-conductor (PEC) boundary edges. PEC is applied on all metal surfaces and the exterior box; the two readout ports are left open (lossless) in this first version, so no resistive element enters and the eigenproblem pencil stays real symmetric. Figure 1 shows the geometry with physical groups color-coded.

3.2 The junction model: a lumped reactive shunt

The Josephson junction is modeled identically in both solvers as a lumped reactive shunt — a parallel $L \parallel C$ element with $L = 14.860 \text{ nH}$ and $C = 5.5 \text{ fF}$ — attached to the four-triangle `lumped_element`

¹The TEAM founding documents are workshop reports and proceedings without resolvable DOIs; we describe the tradition in prose rather than cite an arbitrary secondary source.

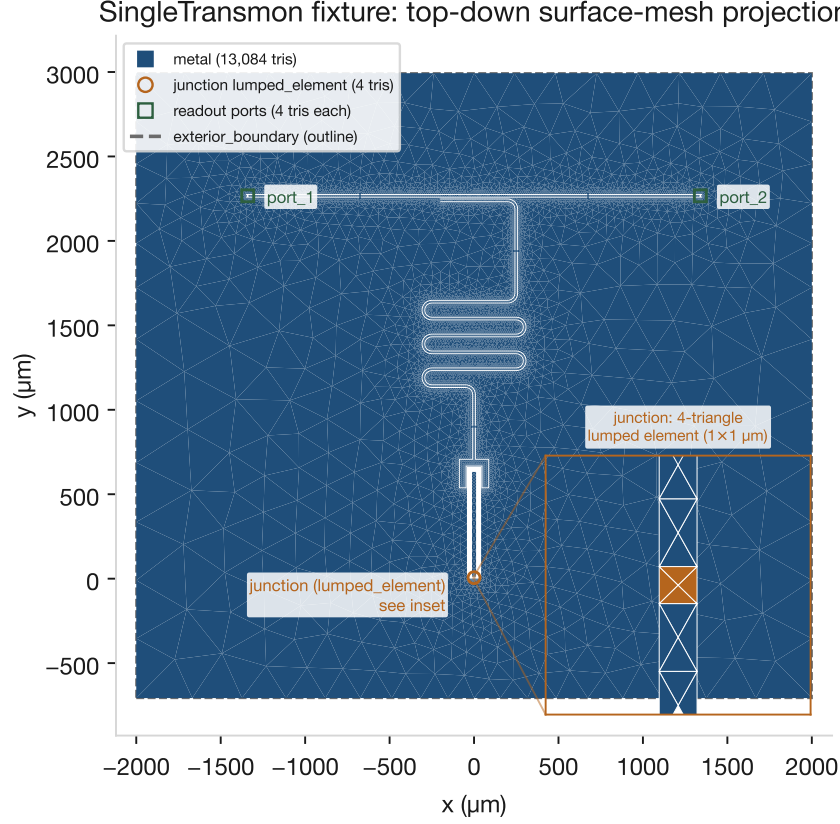


Figure 1: Benchmark geometry: top-down projection of the committed surface mesh (the SHA-256-pinned MSH 4.1 fixture) for the DeviceLayout.jl v1.15.0 `SingleTransmon` transmon and readout resonator. Surface physical groups are color-coded — the metal ground plane (with the coplanar-waveguide feedline, meander resonator, and transmon slot as etched gaps), the four-triangle junction `lumped_element` surface (enlarged in the inset, actual triangulation shown), the two readout ports, and the exterior-boundary outline; the **substrate** and **vacuum** volume groups fill the enclosing box. Both solvers consume this identical mesh (22,684 nodes; 133,314 tetrahedra).

surface group Γ . Following the uniform lumped-port construction, with port length ℓ and width w , the shunt contributes

$$K_{\text{port}} = \frac{\ell}{wL} S_{\Gamma} \quad \text{and} \quad M_{\text{port}} = \frac{C\ell}{w} S_{\Gamma}, \quad (1)$$

where S_{Γ} is the port-surface matrix assembled on Γ . The inductive term is frequency-independent and adds to the stiffness matrix; the capacitive term adds to the mass matrix. The eigenproblem is then the real symmetric generalized pencil

$$(K + K_{\text{port}})x = \omega^2 (M + M_{\text{port}})x. \quad (2)$$

In Palace, the same physics is entered as a reactive `LumpedPort` ($L \parallel C$, no resistance) on the same surface group; the readout ports are configured open (no 50Ω termination), so Palace's pencil is likewise real. This one-to-one correspondence of the junction treatment is what makes the comparison same-physics as well as same-mesh.

Table 1: Solver architectures compared. The two implementations share the discretized problem definition (identical mesh, first-order Nédélec elements, identical lumped-shunt junction) and are otherwise independent.

	geode-fem	Palace
Language / substrate	Rust / Burn tensors (f64)	C++ / MFEM
Assembly	sparse full-tensor Nédélec	MFEM $H(\text{curl})$
Eigensolver	shift-invert Lanczos	SLEPc-class Krylov
Inner linear solve	faer sparse LU (direct)	AMS-preconditioned iterative
Spurious-mode control	none (post-hoc filter; §8)	divergence-free projection
Parallelism	single process	MPI-distributed

4 Solver Architectures Under Comparison

The two solvers share the mathematical problem (Eq. 2) and nothing else: no code, no language, no linear algebra, no eigensolver strategy. Table 1 summarizes the contrast.

geode-fem assembles the full-tensor Nédélec system sparsely in double precision, reduces to interior DOFs, and solves the pencil with a real shift-invert Lanczos iteration whose inner solves are direct sparse-LU factorizations. It runs in a single process with no intra-solve parallelism today. Palace assembles via MFEM, applies a divergence-free projection to confine the Krylov iteration to the physical subspace, preconditions with auxiliary-space methods (AMS) (Hiptmair and Xu, 2007; Koley and Vassilevski, 2009), and distributes over MPI ranks. These choices have consequences for both correctness behavior (Section 8) and performance (Section 9); the point of the benchmark is to expose those consequences honestly rather than to declare a winner.

5 Validation Methodology

The benchmark protocol is oracle-first: every claim is anchored either to an exact analytic quantity or to a committed reference artifact, and the test suite is designed so that silent pipeline failures are caught by construction.

Same-mesh, same-physics protocol. The mesh is generated once, committed, and pinned by SHA-256. Both solvers consume it at matched (first) order with the identical junction model (Section 3.2). This removes meshing and model entry as confounders: any residual disagreement is attributable to formulation or solver implementation.

Analytic tripwires. Two exact quantities guard against coincidental or self-consistent-but-wrong agreement. First, the isolated junction LC resonance $f_{LC} = 1/(2\pi\sqrt{LC}) = 17.60$ GHz anchors the expected location of the junction mode. Second, a Josephson inductance-scaling tripwire: doubling L must move the junction mode by exactly $1/\sqrt{2}$. In our runs the junction mode moves from 17.4901 to 12.37 GHz, a ratio of $0.7071 = 1/\sqrt{2}$ to the reported precision (Figure 3); the measured tripwire values are committed alongside the benchmark results (Section 11).

Inverse tests that must fail. Alongside tests asserting agreement, the suite contains perturbed configurations (such as the doubled- L case) that are *required to disagree* with the committed oracle. A pipeline that still “passes” under such a perturbation is not measuring anything; the inverse tests make that failure mode loud.

Participation-based mode identification. Modes are identified, not eyeballed — but the identification must respect the fact that the two solvers expose *different* participation diagnostics.

Table 2: Eigenmode agreement on the identical mesh (22,684 nodes / 133,314 tets / 156,863 Nédélec DOFs, 133,108 interior after PEC), first order in both solvers, junction as a lumped reactive shunt ($L = 14.860$ nH, $C = 5.5$ fF). geode-fem at commit 3174015; Palace at commit fba6a5b. All values are quoted verbatim from the committed `benchmarks/transmon_eigen/results.toml` (mode keys in the first column): geode-fem frequencies at the committed three decimals, Palace at full precision, and the committed per-mode `rel_err_pct`, which is computed against the rounded geode-fem values (precision disclosure in Section 11).

Mode		geode-fem (GHz)	Palace (GHz)	Δ (%)
<code>resonator</code>	readout resonator	5.153	5.151335830348	0.032
<code>mode_2</code>	box/cavity	15.465	15.46052107794	0.029
<code>junction_lc</code>	junction LC	17.490	17.49010903536	0.001
<code>mode_4</code>	box/cavity	18.693	18.69165792915	0.007
<code>mode_5</code>	box/cavity	20.703	20.69755679425	0.026
<code>mode_6</code>	box/cavity	26.088	26.08089940472	0.027

Within geode-fem, for each eigenvector x the stiffness-participation ratio

$$p = \frac{x^\top K_{\text{port}} x}{x^\top (K + K_{\text{port}}) x} \quad (3)$$

measures the fraction of pencil energy stored in the junction inductor and cleanly separates the junction LC mode ($p = 1.000$) from the resonator and box modes ($p = 0.000$ at the artifact’s three-decimal precision). Palace reports a per-port energy-participation (port-EPR) output, which is a field-based, differently normalized quantity; on the committed run the two diagnostics do *not* rank the modes the same way (in Palace’s committed `port-EPR.csv` the 17.49 GHz junction mode carries the smallest-magnitude port-EPR of the six modes, all of which are $\leq 5 \times 10^{-4}$). The two quantities are complementary diagnostics, not a shared discriminant, and we do not compare them numerically. Cross-solver mode identification therefore rests on *frequency* agreement — the junction mode appears at 17.490 GHz (geode-fem) and 17.4901 GHz (Palace), matching the analytic anchor $f_{LC} = 17.60$ GHz — with geode-fem’s participation used to identify the junction mode on its own side and to expose the spurious mode of Section 8.

Determinism check. The Palace oracle is not a one-off: rerunning Palace from the committed configuration reproduced its committed `eig.csv` bit-for-bit.

6 Results: Cross-Solver Agreement

Table 2 presents the headline result: on the identical 133k-DOF mesh at matched order, all six modes agree to $\leq 0.033\%$ (worst case 0.032%), and the junction LC mode — the mode the junction model exists to produce — agrees to 0.001%.

Three features of Table 2 deserve comment. First, the junction mode sits at 17.490 GHz (geode-fem) and 17.4901 GHz (Palace), 0.6% below the isolated-circuit anchor $f_{LC} = 17.60$ GHz — the small shift reflects the junction’s embedding in the field problem, and its near-coincidence with the analytic value in *both* solvers is the tripwire working as intended. Second, the mode identification is robust: within geode-fem, junction participation is $p = 1.000$ for the 17.49 GHz mode and $p = 0.000$ for every other physical mode, and the cross-solver assignment rests on frequency agreement against the analytic anchor rather than on comparing the two solvers’ differently normalized participation outputs (Section 5). Third, the agreement is between genuinely independent code paths — a direct-factorization Lanczos in Rust and a projected, preconditioned Krylov method in C++ — so

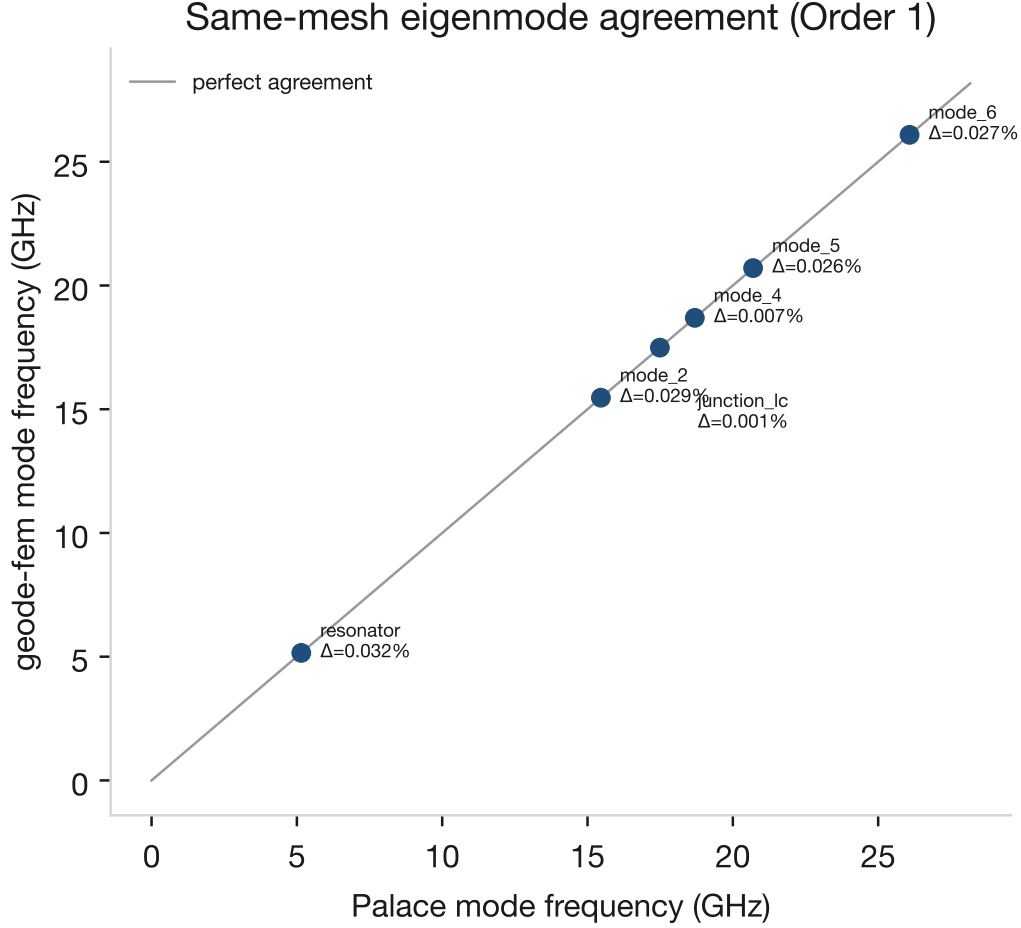


Figure 2: Mode-frequency agreement: geode-fem versus Palace on the identical mesh, with per-mode relative deviations annotated. Data from the committed benchmark artifacts (`benchmarks/transmon_eigen/results.toml` and Palace’s committed `eig.csv`).

a common implementation error is an unlikely explanation for the residual $\leq 0.033\%$ (worst case 0.032%), and the analytic tripwires exclude self-consistent-but-wrong agreement about the junction physics.

7 The Absent Qubit Mode: An Expected Negative

A reader familiar with the DeviceLayout.jl blog workflow (Peairs and Carson, 2025) will notice that its reported spectrum spans $[4.14, 5.591]$ GHz — including a ~ 4 GHz qubit mode — while Table 2 contains no mode below 5.15 GHz. This is not a discrepancy to be minimized; it is a property of the model, and we state it plainly: **no ~ 4 GHz qubit mode exists in this junction model, by construction.**

The physical transmon frequency arises from the Josephson inductance resonating against the *pad (shunt) capacitance* of roughly 80–100 fF (Koch et al., 2007). The v1 lumped-shunt model of Section 3.2 gives the junction only its own 5.5 fF, so the junction’s resonance lands at 17.5 GHz, not near 4 GHz; the pad capacitance exists in the field solution but is not wired across the lumped element. Both solvers agree on the absence: Palace’s projected eigensolve, instructed to hunt at a

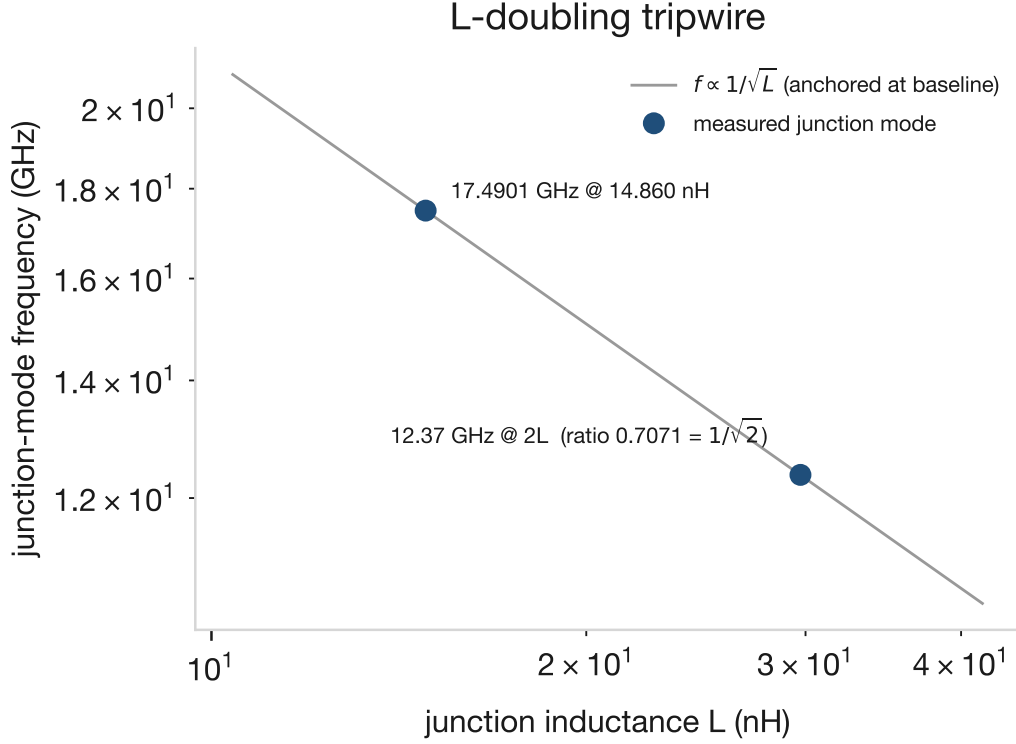


Figure 3: Josephson inductance-scaling tripwire: junction-mode frequency versus junction inductance L on log-log axes. Doubling L moves the junction mode from 17.4901 to 12.37 GHz — ratio $0.7071 = 1/\sqrt{2}$ — exactly as the $f \propto 1/\sqrt{L}$ scaling requires. Measured values from the committed `benchmarks/transmon_eigen/results.toml` tripwire block.

shift of $\sigma = 4.5$ GHz, finds nothing below the 5.15 GHz resonator mode. The blog’s [4.14, 5.591] GHz spectrum is therefore *not reproduced by this model*, and no agreement with it is claimed.

The route to genuine qubit quantities from this benchmark is energy-participation-ratio (EPR) post-processing over the field modes (Minev et al., 2021a; Nigg et al., 2012), in which the junction’s participation in each electromagnetic mode — exactly the quantity in Eq. 3 — feeds the quantization. That analysis is planned future work (Phase C of this benchmark program) and is out of scope for the present cross-validation.

8 A Disclosed Spurious Mode

Honest cross-validation requires reporting where the solvers *disagree*, and there is exactly one such place. geode-fem’s Lanczos iteration operates on the raw pencil of Eq. 2 without a divergence-free projection. On this problem it leaks one spurious mode at 3.4528 GHz, below the physical band, localized on the junction surface with participation $p = 0.9942$ (measured values committed alongside the benchmark results; Section 11). Palace, whose eigensolve projects onto the divergence-free subspace, shows no such mode — its spectrum is clean down through the $\sigma = 4.5$ GHz hunt of Section 7.

We classify this mode as spurious — a gradient-nullspace-adjacent artifact of the un-projected iteration, not a physical resonance — on documented criteria: (i) it is absent from Palace’s projected spectrum on the identical mesh; (ii) it sits far below the analytic junction anchor $f_{LC} = 17.60$ GHz while carrying near-total junction participation, a combination inconsistent with the physical junction

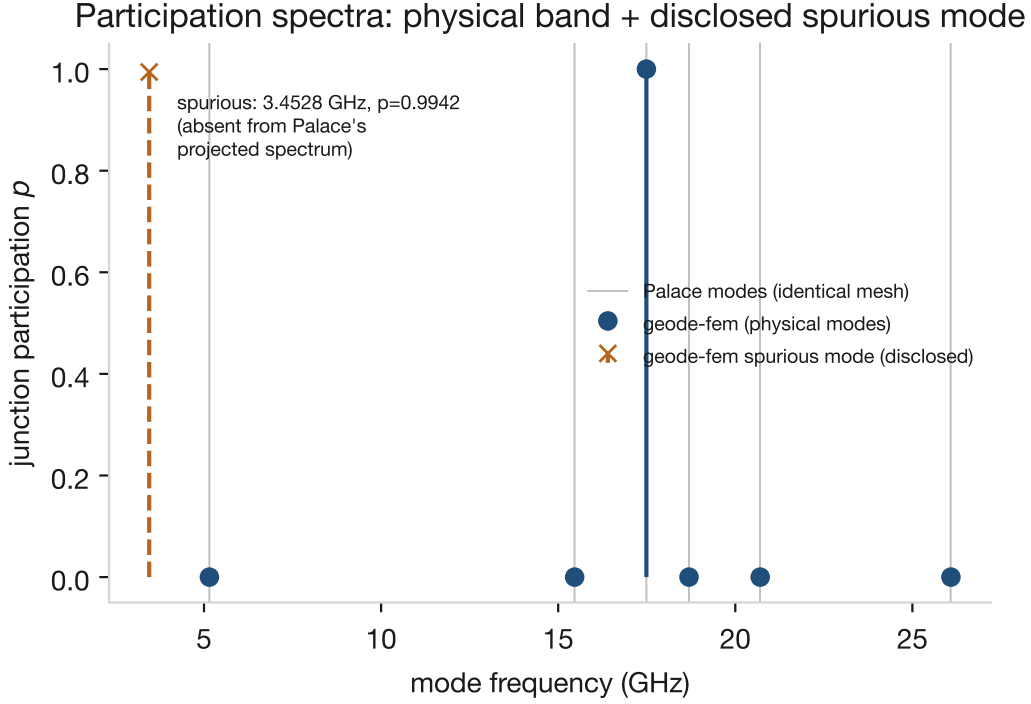


Figure 4: geode-fem junction-participation spectrum on the identical mesh, with Palace’s mode locations (from the committed `eig.csv`) overlaid. The physical junction LC mode appears at 17.49 GHz in both solvers’ spectra (geode-fem stiffness participation $p = 1.000$); geode-fem’s un-projected Lanczos additionally leaks one spurious junction-localized mode at 3.4528 GHz ($p = 0.9942$) with no Palace counterpart. Palace’s field-based port-EPR is a differently normalized diagnostic and is not plotted on geode-fem’s participation axis (Section 5).

mode that appears, in both solvers, at 17.49 GHz; and (iii) it does not respond as a physical mode under the tripwire perturbations. The mode is excluded from Table 2 by these criteria, which are recorded alongside the committed benchmark configuration. Figure 4 contrasts the two solvers’ spectra, spurious mode included.

The fix is known and flagged as future work: a divergence-free or tree-cotree gauge projection on geode-fem’s eigenpath, mirroring the divergence-free projection Palace applies (Palace developers, AWS Center for Quantum Computing, 2023); implementing it is a tracked geode-fem follow-up (repository issue #502). We prefer disclosing the artifact and its filtering criteria over silently post-selecting the spectrum: the disagreement is itself a finding about what the divergence-free projection buys in a production eigensolver.

9 Performance: CPU Cell

9.1 Setup

All CPU measurements ran on a single Amazon EC2 `m6i.4xlarge` instance (8 physical cores / 16 vCPUs, 64 GB RAM) in `us-west-2`.² Each measurement covers the *full pipeline* — mesh load, assembly, solve, and output — timed with `/usr/bin/time`; entries with uncertainties are the mean

²On-demand cost approximately \$0.77/hour at the time of the runs; the entire CPU cell is reproducible for a few dollars.

Table 3: CPU-cell wall-clock and memory, full pipeline (mesh load + assembly + solve + output), `m6i.4xlarge`. Uncertainties are over $n = 3$ runs where shown; the 4-rank Palace row is a single run ($n = 1$). Palace memory is the per-rank maximum single-process RSS, not an aggregate across ranks. Values from the committed `benchmarks/transmon_bench_cpu/results.toml`. Palace at 16 ranks (hyperthreads) was refused by MPI process binding and is excluded — it is not a data point.

Solver	Parallelism	Wall (s)	Peak RSS
geode-fem @3174015	1 process (serial faer sparse-LU shift-invert Lanczos)	51.2 ± 0.4	3.1 GB
Palace @fba6a5b	4 MPI ranks	50.84	0.69 GB/rank
Palace @fba6a5b	8 MPI ranks	30.6 ± 0.1	~ 0.5 GB/rank

over $n = 3$ runs. geode-fem ran at commit 3174015, Palace at commit fba6a5b, both release builds. The complete cell — per-run wall clocks, peak resident-set sizes, and instance/software provenance — is committed as `benchmarks/transmon_bench_cpu/results.toml`.

9.2 Results

Table 3 supports two statements that must be read together. Per core, geode-fem’s serial direct-factorization eigensolve is roughly $4\times$ more efficient than Palace’s distributed Krylov–Schur/AMS solve on this 133k-DOF problem: one geode-fem core matches four Palace ranks almost exactly (51.2 versus 50.84s). At the whole-box level, Palace’s MPI parallelism wins $1.7\times$ (30.6 versus 51.2 s), because geode-fem has no intra-solve parallelism today and leaves seven cores idle.

This is an architecture trade-off, not a ranking. At 133k DOFs the sparse LU factorization fits comfortably in memory (3.1 GB peak) and its flop efficiency dominates; a distributed iterative method carries communication and preconditioner-setup overheads that cannot amortize at this size, but distributes memory (~ 0.5 GB/rank at 8 ranks) and scales to problem sizes and rank counts where a serial direct factorization is not viable at all. The honest summary is that each architecture wins the regime it was designed for, and this benchmark documents where the crossover sits for a production transmon geometry at first order. Palace at 16 ranks on the hyperthreaded vCPUs was refused by MPI process binding; we exclude it rather than report a degraded configuration as if it were a measurement.

10 Performance: GPU Cell (Correctness Established; Performance Deferred)

10.1 Scope and honest framing

geode-fem’s GPU path today accelerates the *driven* (deterministic frequency-sweep) solve — a matrix-free Nédélec matvec feeding an on-device COCG iteration whose per-iteration host–device synchronization budget is $O(1)$ scalars — and *not* the eigensolve that produced the headline comparison of Section 6. A GPU column for the eigenmode benchmark would therefore be misleading; the GPU cell instead targets the driven-solve pipeline. Additionally, the CUDA backend (burn-cuda 0.21) is f32-only — the underlying cubecgl runtime currently disables f64³ — so all GPU results are mixed-precision-qualified and any future performance numbers will be reported with explicit correctness deltas against the f64 CPU path, not as drop-in equivalents.

10.2 Correctness results

On 2026-07-14, geode-fem code executed on a physical GPU for the first time: a single Amazon EC2 `g6e.xlarge` instance ($1\times$ NVIDIA L40S, 46 GB; driver 595.71.05, CUDA 13.2). Two

³Upstream constraint of the burn/cubecgl stack at version 0.21 (Tracel AI, 2024); tracking-issue URL to be added at revision. TODO(operator).

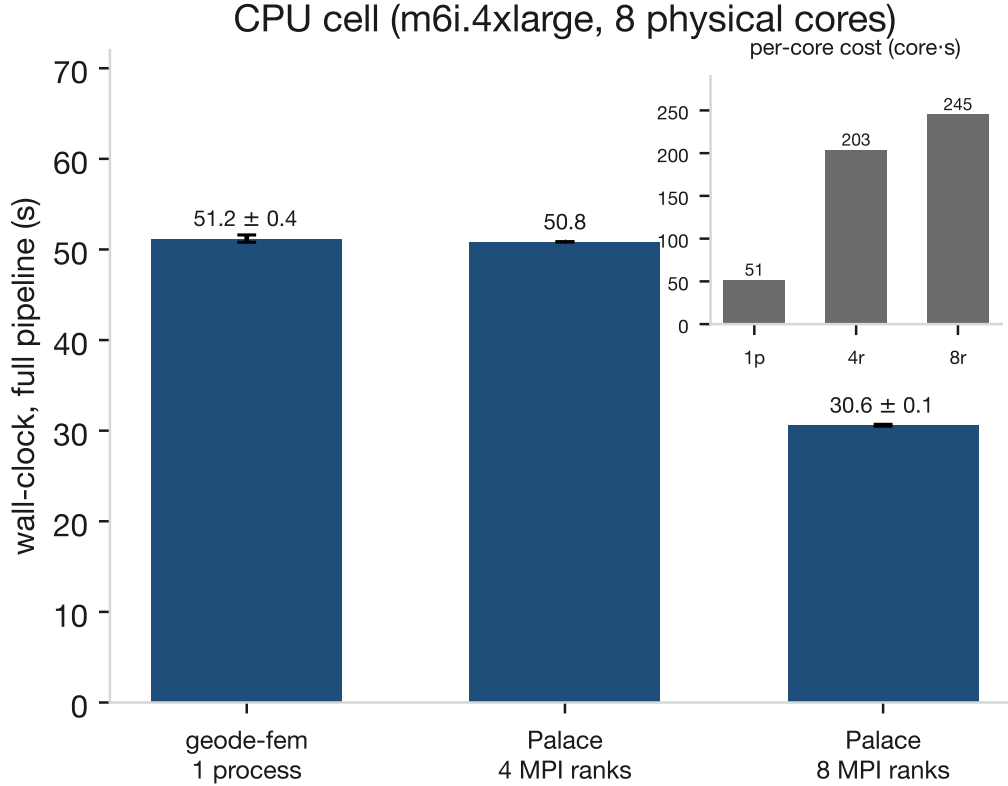


Figure 5: CPU-cell wall-clock comparison (geode-fem single process versus Palace at 4 and 8 MPI ranks), with a per-core-efficiency inset. Full pipeline timings from Table 3.

CUDA-f32 correctness smokes pass on this hardware: the matrix-free Nédélec matvec smoke (`matrix_free_cuda_f32_smoke`, 15.0s wall including GPU initialization and kernel JIT compilation) and the on-device COCG smoke (`cocg_burn_cuda_f32_smoke`, 3.3s). Both tests assert agreement of the CUDA-f32 results with the f64 CPU reference path within f32-appropriate tolerances. The quoted wall times are smoke-test durations dominated by startup cost; they are correctness evidence, not performance measurements.

One gap is disclosed rather than papered over: the end-to-end driven `IterativeMatrixFree` solver path is CUDA-compilable but shipped no runtime smoke on GPU hardware in this measurement campaign. Closing that gap is a tracked geode-fem follow-up (repository issue #499); until it closes, the correctness claim above covers the two constituent kernels (matvec and COCG), not the assembled driven pipeline.

10.3 Performance: explicitly deferred

No GPU performance numbers exist at the time of writing, and none are reported. A driven-solve GPU-versus-CPU scaling benchmark on this fixture family is defined and tracked (repository issue #501); its results — and a Palace-GPU (libCEED/CUDA) cell — will be reported only when they exist as committed artifacts, in a revision or follow-up. The design intent (keeping the Krylov iteration resident on the device with a constant per-iteration synchronization budget) is stated here as architecture, not as a measured speedup.

11 Reproducibility

Every artifact behind every number in this paper is committed and scripted.⁴

- **Geometry and mesh.** Generated by DeviceLayout.jl v1.15.0’s `SingleTransmon` example (AWS Center for Quantum Computing, 2025); the fixture generator is committed, and the MSH 4.1 fixture is pinned by SHA-256 (5b3ff4c3...b33dd; full hash and provenance in the committed `transmon_smoke.provenance.txt`).
- **The MSH conversion gotcha.** MFEM’s MSH 4.1 reader rejects the DeviceLayout multi-entity-block file (“vertices indices are not unique”), so the Palace runs consume an MSH 2.2 conversion (`gmsh -save -format msh2`). The conversion is node-preserving (all 22,684 nodes unchanged, physical groups preserved) and physics-neutral: eigenvalues reproduce bit-for-bit across the conversion. The conversion step is documented in the committed `palace_config.provenance.txt`. We document this because it is exactly the kind of silent pipeline step a same-mesh claim must disclose.
- **Solver configurations.** The Palace configuration and its provenance file are committed under `reference/fixtures/transmon_palace/`; geode-fem’s benchmark definitions live in `benchmarks/transmon_eigen/results.toml`, alongside the exact benchmark commands.
- **Precision of the committed agreement numbers.** `benchmarks/transmon_eigen/results.toml` stores geode-fem frequencies at three decimals while the Palace column carries full precision, so the committed per-mode `rel_err_pct` is computed against the rounded geode-fem values. At full precision the agreement is slightly better (approximately 0.029% worst case). We quote the committed-artifact numbers throughout and disclose the rounding here.
- **Performance and tripwire artifacts.** The complete CPU cell (per-run wall clocks, peak RSS, instance and software provenance) is committed as `benchmarks/transmon_bench_cpu/results.toml`. The measured L-doubling tripwire values (17.4901 $\rightarrow$ 12.37 GHz, ratio 0.7071) and the spurious-mode measurements (3.4528 GHz, $p = 0.9942$) are committed in `benchmarks/transmon_eigen/results.toml`.
- **Oracles.** Palace’s raw outputs (`eig.csv`, per-port participation, run log) are committed; a rerun from the committed configuration reproduced `eig.csv` bit-for-bit.
- **Hardware disclosure.** CPU cell: EC2 `m6i.4xlarge` (8 physical cores / 16 vCPU, 64 GB), `us-west-2`. GPU cell (correctness): EC2 `g6e.xlarge` (1 $\times$ NVIDIA L40S 46 GB; driver 595.71.05, CUDA 13.2).
- **Versions.** geode-fem commit 3174015; Palace commit fba6a5b; DeviceLayout.jl v1.15.0; Burn/burn-cuda 0.21.

12 Discussion

What the agreement does and does not establish. In V&V terms (Oberkampf and Roy, 2010; Roache, 1997), this study is code verification and code-to-code comparison, not validation against nature: two independent implementations agreeing to $\leq 0.033\%$ (worst case 0.032%) on the identical

⁴Repository URL and archival DOI for the geode-fem source at commit 3174015 to be added at submission. TODO(operator).

discrete problem, with analytic tripwires pinning the junction physics. It does not certify that either solver’s answer matches a measured device — that is what Ye et al. (2025)-style experimental benchmarks address, and the two evidence types are complementary. What same-mesh agreement uniquely rules out is the large class of errors that live in formulation, assembly, boundary-condition handling, and eigensolver machinery.

Threats to validity. A common-mode error — both solvers wrong in the same way — cannot be excluded by cross-comparison alone; the analytic anchors (f_{LC} , the $1/\sqrt{2}$ scaling) and the genuinely disjoint implementations make it unlikely for the junction physics specifically. Configuration-entry mismatch (comparing subtly different problems) is mitigated by committed configurations, the shared mesh file, and participation-based mode identification on the geode-fem side with frequency-anchored cross-solver assignment (Section 5). Performance numbers are $n = 3$ full-pipeline measurements on one instance type; they support the architecture-level reading in Section 9, not microsecond-level claims.

Limitations. (i) One geometry, one mesh, first order only — higher-order and mesh-refinement convergence is future work (a $p = 2$ Palace configuration is already committed for that purpose). (ii) The lossless v1 model leaves readout ports open; resistive terminations (and hence complex-symmetric pencils and mode Q s) are out of scope. (iii) The spurious-mode filter of Section 8 currently leans on the committed oracle and documented criteria; the principled fix (divergence-free/tree–cotree projection) is identified and issue-tracked but not yet implemented. (iv) The per-core efficiency result is specific to the 133k-DOF regime where a serial direct factorization is memory-feasible; it will not extrapolate to problem sizes that demand distribution. (v) The GPU cell establishes correctness of the driven-path kernels only; the GPU performance study is deferred, and no GPU performance claims are made in this version.

13 Conclusion

We presented the first independent, quantitative validation of the Palace and DeviceLayout.jl superconducting-transmon simulation stack: an independently implemented Rust FEM solver reproduces Palace’s eigenmodes on the identical mesh and identical junction model to $\leq 0.033\%$ across all six modes (worst case 0.032%; 0.001% on the junction LC mode), with exact analytic tripwires guarding the junction physics. The benchmark’s methodology — same-mesh protocol, oracle-first tripwires, inverse tests that must fail, participation-based mode identification, and honest-negative reporting of both the by-construction absent qubit mode and a disclosed spurious mode — is itself a reusable template for cross-solver validation in this domain. The CPU performance cell documents a clean architecture trade-off: $\sim 4\times$ per-core efficiency for the serial direct-factorization eigensolve against a $1.7\times$ whole-box win for MPI distribution at 133k DOFs. A mixed-precision GPU cell for the driven solve has passed its first correctness smokes on physical hardware; its performance study is deferred until measured artifacts exist. Both solvers emerge with their correctness confirmed on this problem class — which is precisely the outcome an ecosystem that increasingly trusts open solvers for quantum-hardware design should demand, and until now could not cite.

Acknowledgments. geode-fem’s development was AI-orchestrated (agent-driven implementation with human direction and review); this benchmark program and paper follow the same discipline, with every reported number traced to a committed, human-auditable artifact.

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